

DERIVADAS

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$$\frac{d}{dx}(k) = 0; \quad k \text{ es constante}$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(k u) = k u'$$

$$\frac{d}{dx}(x^n) = n x^{n-1}$$

$$\frac{d}{dx}(u^n) = n u^{n-1} \cdot u'$$

$$\frac{d}{dx}(\sqrt{u}) = \frac{u'}{2\sqrt{u}}$$

$$\frac{d}{dx}(\sqrt[n]{u}) = \frac{u'}{n \sqrt[n]{u^{n-1}}}$$

$$\frac{d}{dx}\left(\frac{1}{u}\right) = -\frac{u'}{u^2}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$\frac{d}{dx}(u \pm v) = u' \pm v'$$

$$\frac{d}{dx}(u v) = u'v + uv'$$

$$\frac{d}{dx}(e^u) = e^u \cdot u'$$

$$\frac{d}{dx}(a^u) = a^u \ln a \cdot u'$$

$$\frac{d}{dx}(u^v) = u^v \left(v' \ln u + v \frac{u'}{u} \right)$$

$$\frac{d}{dx}(\ln u) = \frac{u'}{u}$$

$$\frac{d}{dx}(\log_a u) = \frac{u'}{u \ln a}$$

$$\frac{d}{dx}(\sin u) = \cos u \cdot u'$$

$$\frac{d}{dx}(\cos u) = -\sin u \cdot u'$$

$$\frac{d}{dx}(\tan u) = \sec^2 u \cdot u'$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u \cdot u'$$

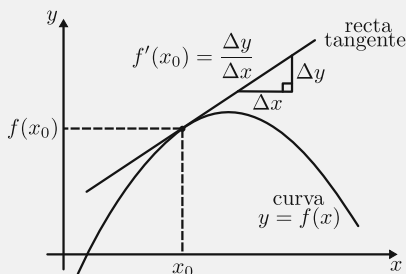
$$\frac{d}{dx}(\sec u) = \sec u \tan u \cdot u'$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u \cdot u'$$

$$\frac{d}{dx}(\arcsen u) = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}(\arccos u) = -\frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}(\arctan u) = \frac{u'}{1+u^2}$$



$$\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$